A Dynamic Simulation Platform for Soft Magnetic Tactile Sensing and Exploration of Model Improvement Directions

Abstract Three-dimensional motion reconstruction with soft magnetic tactile sensors is affected by both spatial magnetic encoding and deformation memory. A small set of regular physical motions cannot cover the resulting dynamic states. We develop a fifteen-channel response surrogate driven by commanded motion and causal history, and integrate continuous three-dimensional interaction, batch trajectory generation, response visualization, frozen-algorithm reconstruction, and ablation experiments in one research platform. A fixed static response is combined with symmetric axis-separated feature increments and multiple history timescales, allowing motion paths and arrival histories to be controlled separately. Matched real and simulated trajectories reproduce the main benefits of slow-history compensation and causal planar learning. A sufficiently trained candidate using 1000 virtual trajectories reduces XY error from 0.423 to 0.226 mm on independent virtual geometries and achieves a 14.8% paired XY improvement on previously collected Yellow-device records excluded from training and checkpoint selection. In a new native online comparison, one complex trajectory improves translation-centered XY error by 29.4% and residual jump rate by 51.4%; another reduces jump rate by 33.0% while incurring a shape-error penalty. A separate ordinary 128-trajectory candidate reduces jump rate on all 36 complex real records across three batches, with mean reductions of 14.5%, 14.1%, and 6.2%, and also reduces dwell fluctuations. In a new virtual-only action-recognition experiment, causal inputs increase held-out-geometry macro-F1 from 0.453 to 0.616, improving 63 of 64 trajectories. These findings support simulation as a means of expanding motion experience and identifying useful directions in device transfer, action-dependent shape accuracy, and temporal stability. Candidates, actions, and metrics are reported separately, with original coordinate bias and accuracy costs retained. The platform is not a validated full physical digital twin, nor does its value require every attempted model to improve simultaneously.

Keywords Soft magnetic tactile sensing, simulation platform, dynamic forward model, causal state, three-dimensional trajectories, virtual data augmentation

I Introduction

A soft magnetic layer encodes contact-induced geometric changes in the responses of fixed Hall sensing nodes. Contact position, indentation, and motion history all affect these responses, so static position calibration cannot describe every dynamic state. Repeated circles or radial scans can test repeatability but provide limited coverage of changing curvature, phase differences between axes, restart after dwell, and alternating tangential and normal motion. We ask whether existing source-device motion records can support a continuously driven forward model that generates motion experience absent from the original training set, and whether the same reconstruction algorithm can turn that experience into transferable benefits.

Physical experiments cost more than acquisition time: mounting, contact-zero calibration, motion execution, and unobserved deformation history also constrain iteration. A continuously driven simulator allows the same path, clock, and noise realization to be reused while varying history strength, dwell, and action order. These controls turn mixed factors into explicit research questions. Simulation does not replace physical conclusions; it accelerates hypothesis formation, rejection, and refinement, and helps select a small set of informative physical experiments from a large action space. Our platform treats magnetic streams with traceable provenance, state resets, and whole-trajectory roles as research records that can enter causal reconstruction and training just like acquisition records.

Simulation also makes dynamic inverse problems easier to inspect. A spatial path alone does not reveal history-dependent responses at the same position or distinguish a stationary indenter from an equilibrated magnetic signal. Linked views of the indenter, five sensing nodes, fifteen response channels, and independent inverse outputs expose these temporal relationships. Visualization supports interpretation and debugging of branch errors, hysteresis,

and depth changes; it is not a substitute for physical validation. Node vectors are not magnetic field lines, and an empirical response animation is not a finite-element displacement field.

Joint mechanical and magnetic modeling has been studied for soft tactile sensor design [1]. Material-point simulation has also been applied to deformation compression and magnetic signal modeling [2]. Optical tactile research shows that diverse simulated contacts can support neural simulation-to-real transfer after appropriate real-data validation [3]. These precedents motivate simulation-based learning, but results from other sensing mechanisms, contact regimes, and materials do not guarantee this device's accuracy.

We use an empirical state-space surrogate for the trajectory task. It predicts the responses of five fixed nodes rather than claiming to recover the complete volume magnetic field or silicone displacement field. Forward fitting uses recorded motion and magnetic responses, not inverse-network coordinates; inverse predictions are therefore not recycled as simulated ground truth. Contributions are a static-equilibrium-preserving axis-separated dynamic representation, an interactive and reproducible research platform, diverse virtual expansion with unchanged inverse architecture, and a controlled chain from weak-condition discovery to training intervention and candidate evaluation. Success does not require all candidates to improve. The main text highlights useful directions while retaining complete controls in supplementary material. Response prediction, mechanism reproduction, simulator-internal learning, and real transfer remain distinct evidence layers.

We additionally use virtual magnetic streams for causal command-action recognition, testing a research task distinct from continuous XYZ regression. Classes follow nominal motion rules; inference uses magnetic observations and timing only. These labels are not independently measured contact or force. This branch is frozen and evaluated separately from trajectory augmentation and does not replace real-transfer validation.

II Scientific Problem and Model Theory

A Coupling between position and deformation memory

Let q be the commanded indenter coordinate, s the latent deformation-history state, and S the five-node observation operator. Equation (1) summarizes the observation. B denotes the contact- and history-dependent spatial field, B0 a confirmed unloaded baseline, and noise includes sampling disturbances and unmodeled effects. Positive nominal Z denotes indentation; extremely shallow commanded motion is not direct ground truth for actual contact depth.

$$\Delta\mathbf{b}_t = S\{\mathbf{B}(\mathbf{r}; q_t, s_t)\} - \mathbf{b}_0 + \boldsymbol{\varepsilon}_t, \qquad q_t = (X_t, Y_t, Z_t). \qquad (1)$$

The same position can carry different history states and therefore different responses. The local first-order change has position and history contributions. A static inverse that explains the second term as displacement can produce direction-dependent errors, reversal tails, or incorrect branches. Causal history estimation separates these contributions; it does not predict future coordinates from a trajectory template.

$$\delta\mathbf{b}_t \simeq J_q \delta q_t + J_s \delta s_t + \boldsymbol{\varepsilon}_t. \qquad (2)$$

This is a conditionally observable inverse problem. Within a valid local branch, suppose admissible history uncertainty spans a response-space subspace H. After removing H, the position Jacobian must retain three independent directions. Equation (3) is a local first-order separation criterion. Practical resolution further requires the smallest singular value multiplied by the target displacement to exceed effective magnetic noise. We use this criterion to explain state constraints, not as a global uniqueness theorem for the entire workspace.

$$\operatorname{rank}(P_{H^\perp} J_q) = 3, \qquad \sigma_{\min}(P_{H^\perp} J_q) > 0. \qquad (3)$$

H represents effective uncertainty under causal constraints. The 660 empirical history coordinates below cannot be interpreted as physical degrees of freedom independently observable from fifteen instantaneous channels. History constraints, the static magnetic atlas, and output permission jointly reduce the feasible solution set. Uncertain states

must remain possible outside the supported workspace, near the noise level, or where position and history responses strongly overlap.

B Nonlinear spatial encoding

Spatial features comprise ten low-order terms and one hundred three-dimensional Gaussian functions. With x=X/6, y=Y/6, and d=max(Z,0)/3, the polynomial terms are 1, x, y, d, x squared, xy, y squared, xd, yd, and d squared. Gaussian centers form a grid with X and Y at -6, -3, 0, 3, and 6 mm and Z at 0.6, 1.4, 2.2, and 3.0 mm.

$$r_j(q) = \exp\left[-\tfrac{1}{2}\sum_{a\in\{X,Y,Z\}}\left(\tfrac{q_a - c_{j,a}}{\sigma_a}\right)^2\right], \qquad \boldsymbol{\sigma} = (2.5, 2.5, 0.65)\ \text{mm}. \qquad (4)$$

All 110 features are multiplied by an indentation factor. Static increments vanish for a nominal unloaded input, while historical residuals can persist. The 0.2 mm scale is an empirical feature parameter, not an identified contact threshold or material constant. Gaussian encoding describes smooth local spatial changes without assuming a linear XYZ-to-field relationship.

$$\phi(q) = \tfrac{Z^+}{Z^+ + 0.2}\,[p(q), r_1(q), \ldots, r_{100}(q)]^T, \qquad Z^+ = \max(Z, 0). \qquad (5)$$

C Symmetric axis separation and static equilibrium

From previous and current positions, construct a=(X(t),Y(t),Z(t-1)) and c=(X(t-1),Y(t-1),Z(t)). Average the feature changes obtained by applying tangential motion first and normal motion first, as in (6). The two increments sum exactly to the total feature change. Pure Z motion has zero tangential increment, and pure XY motion has zero normal increment. For combined motion, cross-effects are shared symmetrically rather than assigned through an arbitrary update order.

$$\delta_{xy,t} = \tfrac{1}{2}[\phi(a_t) - \phi(q_{t-1}) + \phi(q_t) - \phi(c_t)], \qquad \delta_{z,t} = \phi(q_t) - \phi(q_{t-1}) - \delta_{xy,t}. \qquad (6)$$

Each axis-separated increment drives three exponentially decaying states with timescales of 0.3, 1.0, and 4.0 s. States use only present and past motion. A confirmed baseline reset, unknown coordinate, or gap longer than 0.5 s clears them rather than inventing an unknown preceding history.

$$h_{a,k,t} = \rho_{k,t}(h_{a,k,t-1} + \delta_{a,t}), \quad \rho_{k,t} = e^{-\Delta t/\tau_k}, \quad a \in \{xy, z\}. \qquad (7)$$

During dwell, feature increments vanish and history states decay toward a fixed static response. Reversal changes excitation signs, preserving distinct responses for different approaches to the same position. The static and history terms in (8) thus represent equilibrium and transients separately. Dynamic fitting cannot freely shift the frozen static base. These exponential modes are response-surrogate components, not three measured silicone relaxation constants.

$$\widehat{\Delta\mathbf{b}_t} = C_0^T\phi(q_t) + \sum_{k=1}^{3}\left[C_{xy,k}^T h_{xy,k,t} + C_{z,k}^T h_{z,k,t}\right]. \qquad (8)$$

The model is linear in readout coefficients and nonlinear in motion. We freeze the 110-by-15 static coefficient matrix before fitting 660-by-15 historical coefficients. Features are scaled using source-training weighted second moments, and ridge regression constrains the fit. Each complete dynamic record receives equal total weight regardless of frame count; transition-boundary frame weights are halved. Timescales, axis separation, and ridge strength are selected only on source validation records.

$$\widehat{C} = \arg\min_C \sum_i w_i \,\|H_i C - (\Delta\mathbf{b}_i - C_0^T\phi_i)\|_2^2 + \lambda\|D_\phi C\|_F^2. \qquad (9)$$

D-phi is the frozen training feature-scale matrix. The selected historical ridge strength is 0.01 and the feature-scale floor is 0.02. Source validation requires pure-Z and grid errors to stay within 110% of the static reference and retains a static fallback candidate.

D How virtual data can change reconstruction

New motion sequences expand the joint distribution of position and history rather than merely repeat an existing circle. Curvature, speed, reversal, and dwell pair similar positions with different histories, providing constraints that help the inverse network distinguish spatial change from persistent response. Online inputs remain current and past magnetic observations, not the generated action coordinates.

Data quantity cannot remove generator bias. If D is locally Lipschitz within a fixed valid branch, the matched real and simulated inputs satisfy the conditional bound (10). Simulation error can be amplified by local inverse sensitivity. Low field NRMSE or similar mean reconstruction errors therefore do not establish pointwise reproduction.

$$\|D(\widetilde{\mathbf{b}}_{\le t}) - q_t\| \le \|D(\mathbf{b}_{\le t}) - q_t\| + L_D \|\widetilde{\mathbf{b}}_{\le t} - \mathbf{b}_{\le t}\|. \qquad (10)$$

The bound applies only to a fixed-length causal window and the same valid output branch. Rejection and discrete branch changes require separate coverage and failure statistics. We consequently distinguish response prediction, mechanism reproduction, new-motion learning, and real transfer, rather than equating simulator-internal improvement with hardware accuracy.

E Dwell response and causal input representations

Once position is fixed, the static term stays constant but historical magnetic contributions with different timescales remain. Let A(k) be the fifteen-dimensional contribution at dwell onset and r the predicted field minus the same-position static equilibrium response. Equation (11) shows that individually decaying modes need not produce a monotonically decreasing residual norm. Contributions can initially cancel and subsequently lose that cancellation as they decay at different rates.

$$r(s) = \sum_k A_k e^{-s/\tau_k}, \qquad \frac{d\|r(s)\|^2}{ds} = -2\, r(s)^T \sum_k \frac{A_k}{\tau_k} e^{-s/\tau_k}. \qquad (11)$$

The derivative sign depends on a vector inner product, not solely on positive timescales. With a single timescale, a nonzero residual norm decays as one exponential. A stopped indenter and a static observation are therefore different conditions; a universal waiting time need not be equivalent across histories. This derivation describes the frozen surrogate, not increasing physical silicone energy or force, and not an identified material law.

To test whether this understanding helps inverse learning, we form exponential averages m from current and past magnetic fields and use their differences from the current field as causal features. Neither motion coordinates, velocity, nor generator internal states enter these features.

$$m_{\tau,t} = \rho_{\tau,t} m_{\tau,t-1} + (1 - \rho_{\tau,t}) \Delta\mathbf{b}_t, \qquad u_{\tau,t} = \Delta\mathbf{b}_t - m_{\tau,t}, \quad \rho_{\tau,t} = e^{-\Delta t/\tau}. \qquad (12)$$

Instantaneous, 0.3 s history, and 0.3/1/4 s history representations use the same nominal network size and initialization. Extra history may reduce ambiguity but can introduce redundancy, noise, or inappropriate long-lived states. Multiscale superiority is a falsifiable hypothesis, not a consequence automatically implied by a three-timescale generator.

III Simulation Platform and Virtual Data

A Physical data source and continuous interfaces

Inputs originate from complete motion and fifteen-channel magnetic streams recorded on a three-axis loading platform. Five MLX90393 nodes under the flexible magnetic layer use four corner positions and an offset center; they are read through a CH341 interface. Host software archives command events, unloaded baselines, and configuration snapshots. Device structure and operating photographs are available in the project's hardware and system media. Record metadata define indenter, magnetic layer, and mounting conditions. Command-derived nominal motion is not encoder position. Appendix A specifies the source data converted into the continuously driven surrogate.

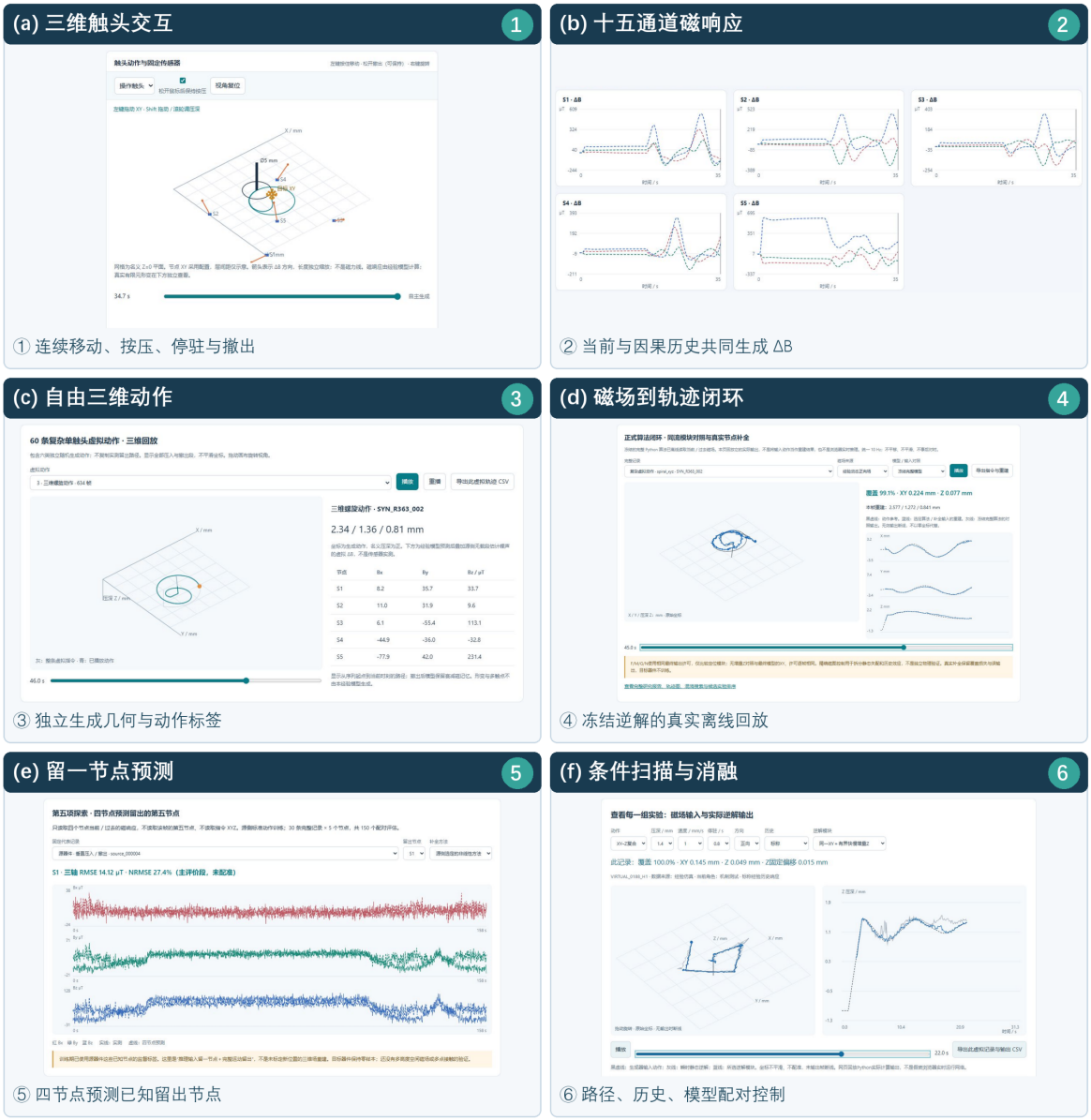


Figure 1. Six actual running-interface panels: (1) continuous 3D motion, indentation, and withdrawal; (2) five-node, fifteen-channel response; (3) independently generated free XYZ motion; (4) virtual streams passed to a frozen inverse and replay of its computed output; (5) four-node prediction of a known held-out node; and (6) paired conditions and inverse-module ablations. Panels retain the original interface language; numbering and captions identify their functions. Example interface values are not pooled into paper statistics. Node vectors are not field lines or finite-element displacements, and generated motion is not mislabeled as reconstructed motion.

The platform supports continuous manual XY movement, indentation adjustment, hold and withdrawal, automatic path replay, and provenance-bearing export. Three-axis readings at five nodes update with motion; paths and

coordinate sequences can be inspected together. Complete Python inverse outputs are saved independently and displayed by the website, which does not substitute input paths for reconstruction. A finite-element viewer can receive external mesh and displacement files but did not generate this paper's data.

B Experiment organization and motion expansion

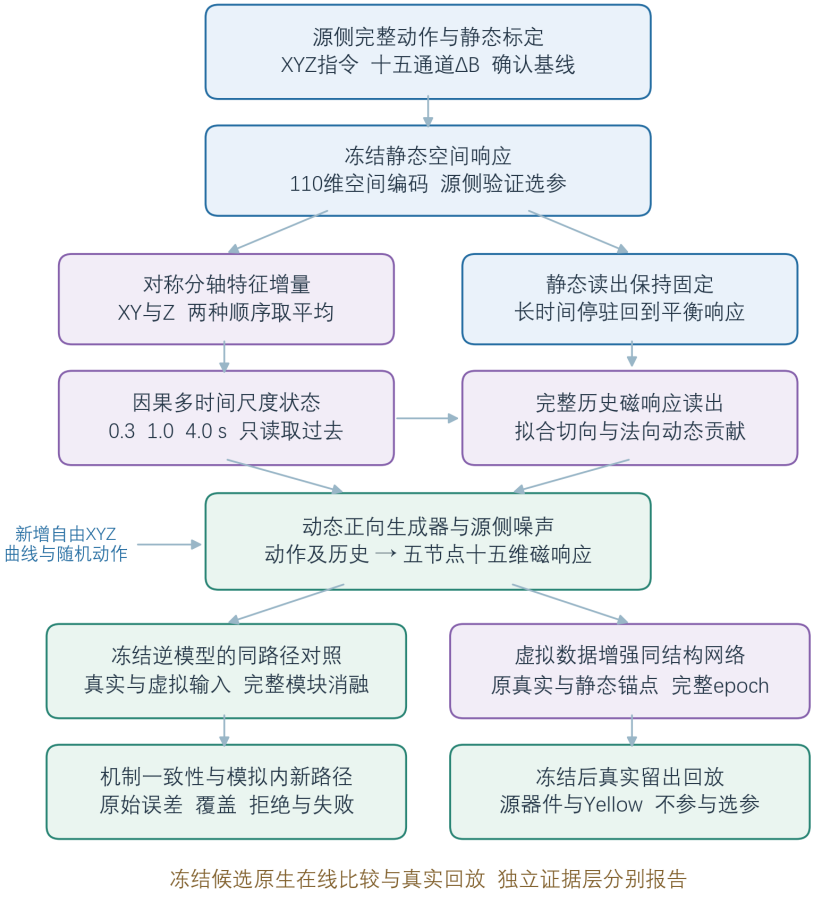


Figure 2. Unified forward-model, virtual-generation, and inverse-learning workflow. Source data determine coefficients, normalization, and training checkpoints. Target devices and declared held-out real paths are evaluated after freezing. Generated fifteen-channel responses and coordinate supervision enter the reconstruction pipeline separately. Newly acquired online comparisons and frozen retrospective replay are separate evidence layers, not pooled averages. The original Chinese diagram is retained with this English explanation of its complete scientific flow.

Four constructions generate spatial Bezier curves, inertial random walks, free 3D waypoints, and multiscale curves. Every complete path has an independent geometry seed; paths are not copies of recorded motions. Nominal contact generation uses X and Y from -5 to 5 mm, Z from 0.45 to 2.4 mm, speed from 0.5 to 2 mm/s, dwell from 0 to 2.5 s, and 10 Hz sampling, with indentation and withdrawal segments retained. Source fitting covers approximately 0.40-2.30 mm indentation, so the upper generated edge is not a physically validated coverage guarantee.

Training expands predetermined prefixes of 100, 500, and 1000 paths. The independent virtual test has 64 complete paths with nonoverlapping geometry seeds. Source unloaded records estimate added channel noise. This scale study does not use domain randomization, and 1000 paths from one empirical generator are not 1000 independent devices.

IV Learning and Evaluation

A Augmentation with unchanged inverse architecture

The baseline is a sufficiently trained, frozen complete causal reconstruction algorithm. Its planar head is a 53-64-64-2 tanh residual network and its depth head a bounded incremental network. Magnetic atlas, slow state, branch rules, shallow-response handling, and output permission form the full chain. Augmentation updates only the planar head, retaining baseline initialization and normalization. Depth and permission parameters remain fixed, but their outputs can change numerically because they read candidate coordinates.

Training retains 7217 source dynamic samples and 565 static anchors. Virtual features are captured through the actual causal chain. Total loss mass is 0.4 for real dynamics, 0.4 for virtual dynamics, and 0.2 for anchors; virtual trajectories are weighted equally to avoid dominance by long records. Dynamic labels are commanded XY minus the frozen branch base, clipped to +/-3 mm per axis. Static residual labels are zero. Motion labels supervise offline training only; online inputs contain causal magnetic observations and base state.

$$\mathcal{L}(\theta)=0.4\,\mathbb{E}_R\|N_\theta(u)-r\|^2+0.4\,\mathbb{E}_V\|N_\theta(u)-r\|^2+0.2\,\mathbb{E}_A\|N_\theta(u)\|^2+\frac{0.001}{2}\sum_l\|W_l\|_F^2. \qquad (13)$$

Adam uses beta1=0.9, beta2=0.999, epsilon=1e-8, batches of at most 256, and seed 7. Each epoch shuffles without replacement and traverses every training sample. Learning rate starts at 0.0002 and halves after 50 complete epochs without relative validation improvement of 1e-4, down to 1e-6. A further 50-epoch plateau at the minimum rate ends training. Each epoch evaluates the original three complete source-validation records; their equal-Trial raw XY error selects the checkpoint. Completion of this protocol is not a claim of global optimality for a nonconvex objective.

Virtual paths	Virtual samples	Samples per full epoch	Epochs	Best epoch	Updates
100	44,542	52,324	911	100	186,755
500	222,231	230,013	659	86	592,441
1000	446,857	454,639	933	143	1,657,008

Table I. Actual complete-traversal training. All groups start from the same sufficiently trained weights and stop after minimum-rate plateau, with eight rate reductions. Each includes the same 7782 real and anchor samples per epoch. Samples are not independent motions. Optimization counts vary with dataset size, so this is a sufficiently trained scale study, not an equal-compute causal test of sample count alone.

B Evidence layers and error definitions

First, complete real records test fifteen-channel prediction after generator freezing. NRMSE normalizes by each reference response's RMS scale. Second, identical command paths and times provide real and generated responses to the same frozen inverse for pointwise and module comparisons. Third, expanded virtual motions train the planar head, which is then tested on independent virtual geometries and previously collected real holdouts. Fourth, baseline and augmented models are acquired online in separate runs of the same protocol; the two complex conditions and controls are reported here. Frozen candidates also process identical raw real streams, actual B0, dt, and event order to test temporal stability. Input-contract verification, not an acquisition label alone, supports replay equivalence.

The earlier real replay has 18 records: seven contact motions and two unloaded controls per source and target device. Motions include two depth grids, pure Z, eight directions, a circle, a compound motion, and a 29-waypoint 3D path. They are excluded from augmentation training and checkpoint selection but were previously analyzed in the project. This is frozen retrospective holdout replay, not a new blind test.

Primary metrics are raw XY and XYZ Euclidean error and absolute Z error, averaged within each complete contact Trial and then equally across Trials. Coverage and erroneous safe-withdrawal outputs are separate; rejected outputs do not count as zero error. Common-valid-frame comparisons diagnose acceptance-mask effects. Real reference

paths use command timing and estimated travel duration, not encoders or independent contact metrology. Main curves receive no post hoc clock shift, coordinate registration, or output smoothing.

V Experimental Results

A Dynamic response and matched trajectories

On the latest source and target contact records, generator magnetic NRMSE is 25.10% and 54.35%. Sending matched real and simulated responses to the same frozen inverse yields source XYZ errors of 0.286 and 0.257 mm and target errors of 0.781 and 0.263 mm. Corresponding real-versus-virtual reconstructed outputs still differ by means of 0.348 mm and 0.786 mm. The loop works, but these differences expose device-distribution bias.

Motion source	Field NRMSE	Real XYZ mm	Virtual XYZ mm	Real / virtual coverage
Source device	25.10%	0.286	0.257	99.56% / 100.00%
Target device	54.35%	0.781	0.263	97.99% / 100.00%

Table II. Matched-path reconstruction. All virtual responses use the source generator. The target row reuses target motion, not a target digital twin. Trial-equal errors use each model's accepted frames and list coverage separately. Similar average error is different from pointwise agreement.

B Reproducing problem-solving mechanisms

For seven source contact actions on strict common-valid frames, slow magnetic state lowers mean XY error by 0.187 mm for real input and 0.192 mm for virtual input, with matching change direction in six actions. Adding the causal planar network above a branch-protected base lowers error by 0.435 and 0.387 mm, with matching direction in all seven. Simulation supports two specific mechanisms: static inversion mistakes persistent response for position change, and causal residual learning exploits spatial and historical context. It does not prove complete physical reproduction of every module.

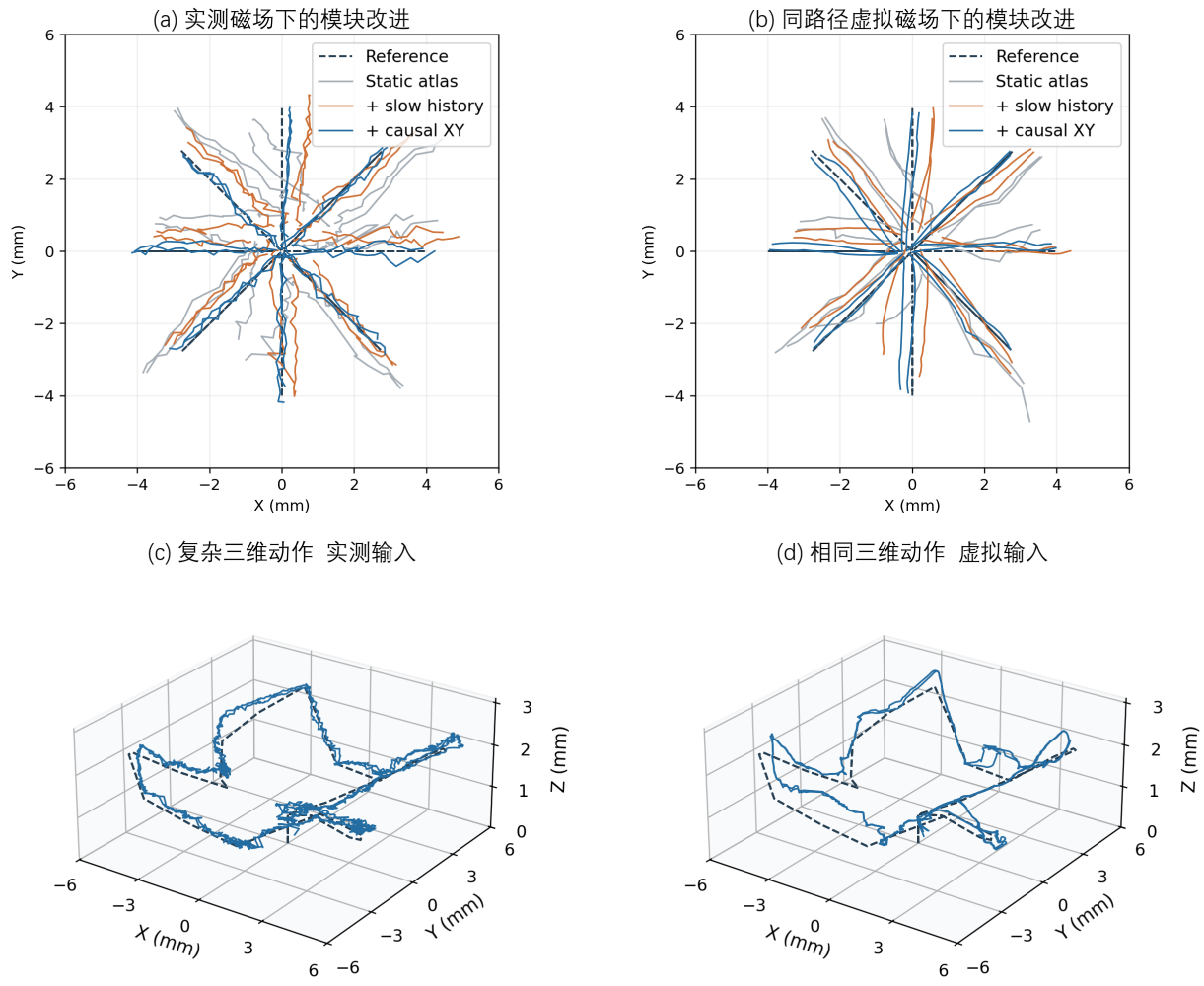


Figure 3. Matched real/virtual reconstruction and module comparisons. The top row uses the fixed first source radial action: static atlas, slow magnetic state, and causal planar network. The bottom row uses a fixed complex source action under real and virtual inputs. Curves show each method's valid outputs, with gaps at rejection. Statistics use all stages' and both inputs' common valid frames. The 3D views share limits; displayed Z scaling is approximately 1.9 times XY without changing coordinates or errors. No coordinate or time registration is used.

Added mechanism	Metric	Real delta mm	Virtual delta mm	Same direction
Slow magnetic state	XY	-0.18682	-0.19250	6/7
Branch protection	XY	-0.00001	-0.00448	5/7
Causal XY network	XY	-0.43507	-0.38682	7/7
Full depth integration	Z	-0.01951	-0.04239	6/7
Bounded Z increment	Z	-0.00956	-0.00251	4/7

Table III. Complete source mechanism comparison. Delta is error after minus before addition; negative means improvement. Average branch-protection gain is small. Bounded-depth increments agree in only four actions and are not primary evidence for universal reproduction. Full depth integration changes state lifecycle and candidate use, not just one neural component. Switching only the bounded Z increment leaves XY and permission identical frame by frame.

C Learning from diverse new trajectories

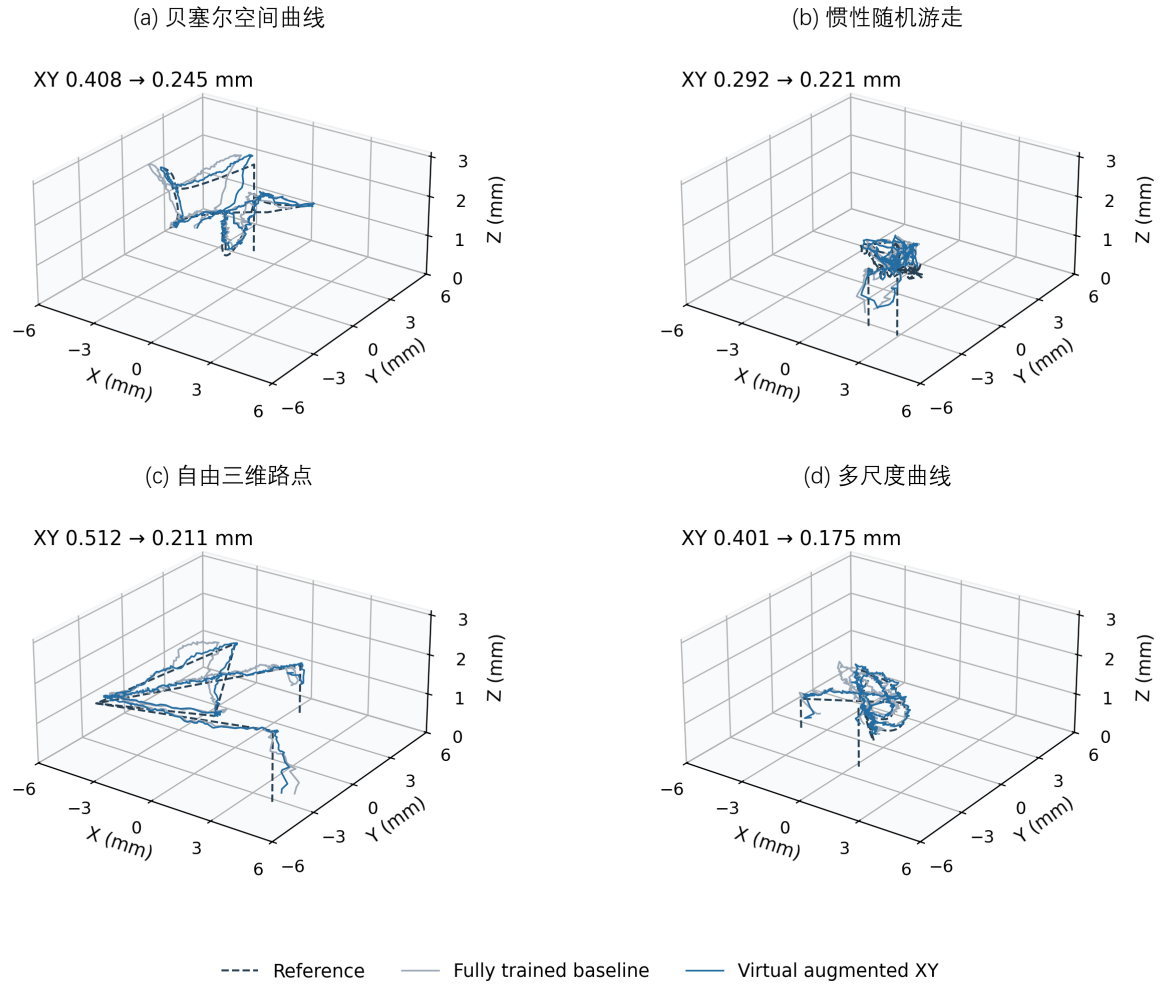


Figure 4. Reconstruction after 1000-path augmentation for four independent virtual test families, always using the first test path in each family. Gray is the sufficiently trained baseline, blue is the full chain with only its planar head updated, and dashed lines are generated reference motions. Original coordinates and Figure 3's displayed Z scale are retained. The depth head is not retrained and rejection gaps remain.

Baseline XY error on 64 independent virtual paths is 0.423 mm. Adding 100, 500, and 1000 diverse motions reduces it to 0.244, 0.233, and 0.226 mm. For 1000 paths, XYZ error falls from 0.460 to 0.277 mm, while Z error changes only from 0.136 to 0.131 mm. Most benefit is planar dynamic reconstruction, not simultaneous improvement of every depth mechanism.

Model	Independent virtual XY mm	Source real replay XY mm	Target real replay XY mm
Fully trained baseline	0.423	0.258	0.738
Add 100 virtual paths	0.244 (-42.2%)	0.303 (+17.4%)	0.678 (-8.2%)
Add 500 virtual paths	0.233 (-45.0%)	0.299 (+15.6%)	0.652 (-11.7%)
Add 1000 virtual paths	0.226 (-46.6%)	0.340 (+31.8%)	0.630 (-14.7%)

Table IV. New virtual motion and raw XY error; parentheses show change from the trained baseline. This scale table uses each model's own valid frames. Virtual testing comprises 64 independent paths; real testing has seven contact records per device. Source training/validation alone selects checkpoints; target results do not select scale or parameters. A further common-valid-frame check gives Yellow baseline 0.73968 mm and candidate 0.62990 mm, a 14.84% improvement. The 14.67% own-mask value and 14.84% common-mask value are different estimands, not separate experiments.

Target-device own-mask XY error improves by 8.2%, 11.7%, and 14.7% with scale; the 1000-path XYZ error improves from 0.781 to 0.677 mm. Common-frame checking confirms a 14.8% raw XY benefit for this candidate, and translation-centered error changes from 0.588 to 0.515 mm. Virtual experience thus changes inversion on real responses without target supervision or selection. Source raw XY error worsens, so this is a useful target-transfer direction, not universal device superiority.

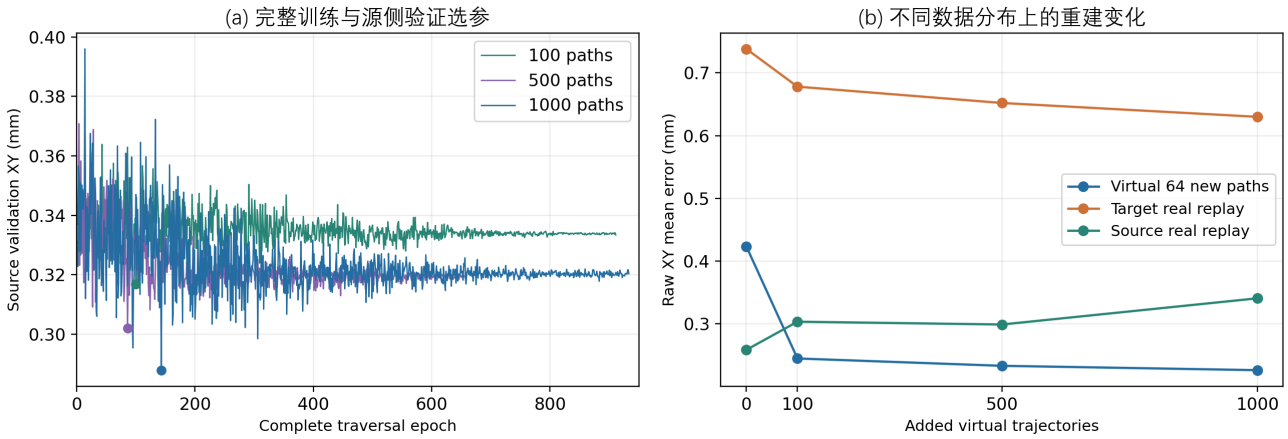


Figure 5. Complete-traversal learning and transfer. Left: every epoch's source validation and selected checkpoint, not a truncated early-training segment. Right: Table IV's own-mask raw XY errors for independent virtual paths and the two real devices. Coverage expansion has clear simulator-internal benefits but different effects across real distributions. Both dataset size and total optimization vary, so sample count alone does not explain the trend.

The 1000-path candidate's mean signed source bias increases relative to baseline by approximately (+0.151,+0.118,+0.003) mm. Removing one constant translation per source record diagnoses about 4.2% better planar shape, but the primary raw-error increase remains. Six of seven target actions improve raw XY; one grid worsens, and pure Z accepts two fewer primary evaluation frames.

Evaluation	Baseline to augmented coverage	Baseline to augmented Z MAE mm	Lost / added frames	Safe outputs / reference frames
Source real replay	99.56% to 99.56%	0.088 to 0.087	0 / 0	0 to 0 / 2,470
Target real replay	97.99% to 97.70%	0.199 to 0.202	2 / 0	0 to 0 / 2,480
Independent virtual paths	99.61% to 99.58%	0.136 to 0.131	7 / 0	359 to 361 / 3,378

Table V. Full-chain checks for 1000-path augmentation. Coverage is a complete-contact-Trial equal-weight mean. Real safe-withdrawal frames have no outputs, but simulated historical tails still trigger output. Primary-segment gains do not hide this failure. Unchanged parameters do not imply unchanged downstream numerical behavior.

D New information from virtual experiments

New information first appears as expanded motion combinations, not more files alone. Defining coupled motion by nominal XY speed and absolute Z speed both exceeding 0.1 mm/s, only 103 of 7965 moving intervals from 24 source generator-training records qualify (1.29%). For 24 free virtual candidates, 4554 of 9181 qualify (49.60%). Equal-trajectory means are 12.90% and 52.51%. Both estimands are retained: pooled moving-time coverage and a typical complete-record fraction. Free 3D motion provides joint excitation scarce in the recorded set, without guaranteeing physical validation of every generated state.

Dwell endpoint searches also yield falsifiable historical-response candidates. The expected real peak is not validated; the complete search and negative result are retained in Appendix D. They are not counted as a positive physical contribution.

E New causal readouts trained with virtual data

Using the frozen generator, 32 complete paths train, 16 independent paths validate, and 64 independent geometries test the representations in (12). All three networks are 60-32-3 tanh readouts with 2051 nominal parameters and identical seed-7 initialization arrays. Current field occupies the first fifteen slots; unused history slots are zero. Active input dimensions are 15, 30, and 60. Zero-slot connections receive no data gradient, so equal nominal size is not proof of equal effective capacity. Input normalization uses virtual TRAIN only; XYZ labels use fixed 5, 5, and 2 mm scales. No real records or internal generator histories enter training.

The 32 paths provide 14518 trainable frames. Every epoch traverses all frames without replacement, minimizing trajectory-equal XYZ squared loss. Sixteen-path raw XYZ validation selects checkpoints. Adam uses initial rate 0.001, batch 512, and weight L2 coefficient 0.0001. A declared 1800-epoch ceiling is combined with 50-epoch plateaus, relative improvement 1e-4, halving, and minimum rate 1e-6 followed by 50 more plateau epochs. All groups stop before the ceiling. Tests open after all three freeze. These direct XYZ research readouts are not the full permission algorithm or production replacements.

The 64 virtual test geometries were previously generated and analyzed. They do not overlap this training or validation set and are tested after freezing, but are not newly blinded geometries. Test outcomes do not select architecture or parameters. Conditions are fixed before execution: moving speed above 0.05 mm/s; dwell below 0.02 mm/s; turning when adjacent moving-direction cosine is below 0.5, followed by a 1 s window; coupled motion when XY speed exceeds 0.05 and absolute Z speed exceeds 0.03 mm/s. These readout diagnostics differ intentionally from Section V-D's 0.1 mm/s coverage threshold.

Input representation	Full epochs	Best epoch	Updates	Rate reductions
Instantaneous field	1694	906	49,126	10
Add 0.3 s history	1485	841	43,065	10
Add three history timescales	1398	818	40,542	10

Table VI. Actual causal-readout training. Every epoch traverses 14518 frames. All groups reach rate 0.000001 and then 50 plateau epochs. Seed, initialization, nominal network size, data roles, and stopping rules are matched. Only virtual validation selects the best epoch.

Representation	XYZ mm and change	XY mm	Z MAE mm	Post-turn XYZ mm
Instantaneous field	0.3026 (+0.00%)	0.2907	0.0593	0.3264
Add 0.3 s history	0.2875 (-4.97%)	0.2765	0.0556	0.2987
Add three timescales	0.3012 (-0.44%)	0.2924	0.0506	0.3313

Table VII. Raw errors on 64 independent virtual geometries; XYZ change is relative to instantaneous input. Post-turn evaluation covers 59 paths containing the event. Short history improves overall XYZ by 4.97%, on 43 paths; multiscale history improves it by 0.44%, on 39 paths. There is no contact-permission gate. Evaluation concerns supported nominal contact frames, not safety coverage or real localization capability.

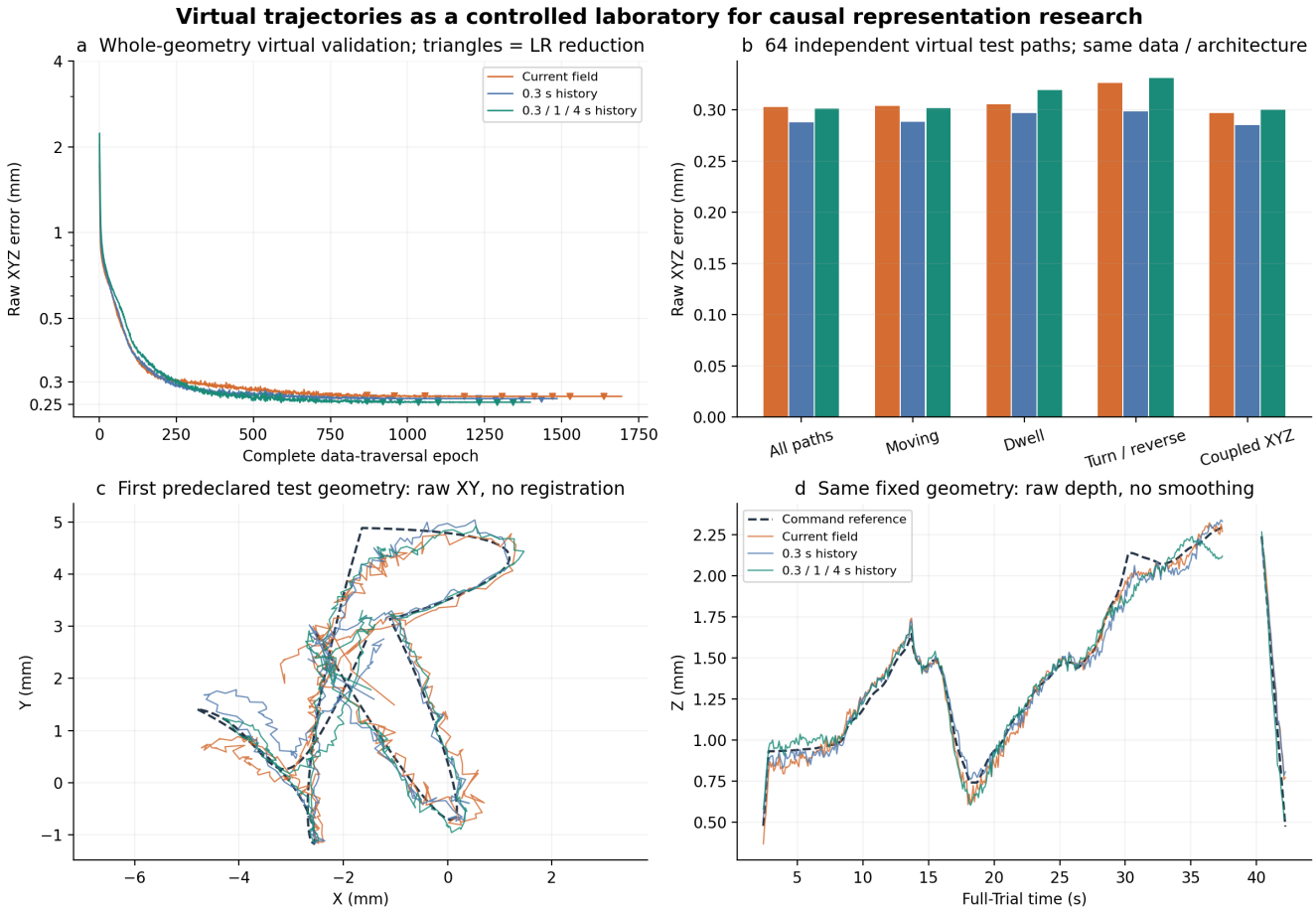


Figure 6. Matched nominal-size causal representation ablation: complete training curves, condition-level results, and raw XY/Z traces for a predetermined test path. Only usable history slots change; effective dimensions are 15, 30, and 60. Whole geometries are held out and every representation is reported. Moving, dwell, post-turn, and coupled-motion metrics are not independent hardware repetitions.

Short history reduces post-turn XYZ error from 0.3264 to 0.2987 mm (8.48%), supporting its use after fast motion in this virtual distribution. Multiscale history reduces Z MAE from 0.0593 to 0.0506 mm (14.64%) but increases XY from 0.2907 to 0.2924 mm and dwell XYZ from 0.3052 to 0.3191 mm. More history is not invariably better. History representations have different tradeoffs for rapidly changing planar localization and depth. Axis-separated or selectively used history is a future test, not an architecture to select on this test and then call independently validated. These readouts have not demonstrated real transfer.

F Useful candidates on new real trajectories

1000-path candidate: joint shape and continuity benefit. The baseline is frozen R373; the candidate uses the 1000-path XY head with original depth and complete-chain parameters. Two same-protocol acquisition plans each complete six Trials: two complex-history paths, two direct-indentation controls, and before/after unloaded records. Each action Trial contains three nested repetitions. Models are acquired online in separate runs, not counterfactuals on the same field stream, so batch and model effects cannot be fully separated. Saved moving outputs are unchanged; both paths have 100% coverage.

Complex path	Raw XY baseline to augmented mm	Centered XY baseline to augmented mm	Residual jump rate baseline to augmented
Condition 001	0.521 to 0.651	0.267 to 0.189	3.91% to 1.90%
Condition 008	0.389 to 0.568	0.159 to 0.175	5.75% to 3.85%

Table VIII. Native online new-path comparison. Condition 001 improves centered XY by 29.4% and jump rate by 51.4%; condition 008 improves jump rate by 33.0% but increases centered error by 10.2%. Raw errors increase in both. Full bias vectors, Z, dwell, and unloaded controls remain in per-Trial supplements. A jump is an XY residual increment above 0.2 mm between consecutive valid frames. Denominators are 3757/3890 and 2959/3483 frame pairs; rejection gaps are not bridged. One translation is fitted over each entire moving segment for shape diagnosis only, not deployed calibration.

This is an action-dependent real improvement direction: the first new complex path improves both shape accuracy and continuity after diagnosing constant coordinate bias. The second improves continuity only, with its shape cost retained. Both complete conditions are reported; one favorable path is not extrapolated to all unknown motions.

Ordinary 128-path candidate: temporal stability. This separately trained candidate uses different weights from the 1000-path candidate. Three batches contain 48 real records, including 36 complex actions: six path types at two speeds per batch. SOURCE and SOURCE repeat use the same device; Yellow is another device. After freezing, all candidates process the same actual stream, B0, dt, events, and complete chain. Paired accuracy uses four-model common-valid frames, while coverage uses each model's mask. SOURCE and Yellow verification covers 32 Trials and 97522 online rows with zero missing rows or validity disagreements and maximum axis difference 2.13e-14 mm. The repeat batch is checked against prior frozen replay, not an identical-model online CSV.

Batch	XY residual jump rate baseline to candidate	Relative reduction	Dwell XY RMS baseline to candidate mm
SOURCE	23.23% to 19.87%	14.47%	0.11195 to 0.10460
SOURCE repeat	24.56% to 21.09%	14.15%	0.11706 to 0.10685
Yellow	46.34% to 43.48%	6.16%	0.15963 to 0.15211

Table IX. Jump rates decrease on all 36 complex records, and dwell fluctuations decrease in all batches. Statistics are equal-Trial means, not pooled frame rates. Raw XY errors change from 0.436 to 0.575, 0.452 to 0.583, and 0.521 to 0.679 mm; centered errors change from 0.215 to 0.253, 0.224 to 0.264, and 0.478 to 0.504 mm. Accuracy costs and stability gains are different quantities; comparing their percentages does not automatically establish overall superiority. All candidates, coverage, jump denominators, and unloaded outputs remain in the source index.

The positive finding is that virtual motion experience can change temporal behavior on real inputs, reducing residual jumps without adding output smoothing. Broader mapping constraints may promote temporal consistency, but training has no explicit adjacent-output regularizer and network Jacobian changes were not independently measured. The explanation remains a hypothesis, not an identified unique mechanism. Localization accuracy is not the only meaningful outcome of this direction.

G Causal command-action learning from virtual magnetic streams

Virtual environments provide reproducible temporal classes as well as coordinate labels. With the frozen source generator and noise distribution, 128 whole geometries train the classifiers, 32 new geometries validate them, and 64 new geometries are generated and tested only after both models freeze. Four construction families remain within one supported regime. Whole geometries are held out, not action classes, devices, tips, or contact types. These are newly initialized classifiers, not fine-tuned R373 coordinate heads; real and Yellow records do not select them.

Five classes are nominal unloaded, hold, XY motion, Z motion, and compound XYZ motion. Commanded Z at or below -0.5 mm defines the unloaded reference. Contact-related labels require generator support and Z from 0.4 to 3 mm. Backward command XY speed and absolute Z speed at or below 0.03 mm/s define stationary axes; values above 0.05 mm/s define moving axes. Intermediate deadbands, shallow states, resets, and first observations after gaps above 0.25 s are unscored. These thresholds define a virtual task, not measured contact thresholds.

The instantaneous model receives the current fifteen channels. The causal model concatenates that field, three exponential averages of the form in (12), and a backward magnetic derivative, totaling 75 features. Its averaging timescales are 0.1, 0.4, and 1.0 s, distinct from the generator's 0.3/1/4 s. At resets or gaps above 0.25 s, averages initialize to the current field and the derivative to zero. Inference receives no coordinates, command velocity, action ID, generator state, or future samples, and adds no output smoothing. Networks are 15-64-64-5 and 75-64-64-5 tanh classifiers. Their input widths and parameter counts differ; this is not an equal-capacity history-only ablation.

Each full epoch visits 57,740 frames. Whole-geometry-equal weights are multiplied by inverse TRAIN-only class mass; the loss is weighted cross entropy plus weight L2 of 0.0005. Normalization uses virtual TRAIN only. Optimizer seed is 7 and layer initialization seeds are 41071, 41072, and 41073; batch size is 256. Learning rate starts at 0.0002 and halves after 50 epochs without relative validation cross-entropy improvement of 0.0001, down to 0.000001. Another 50-epoch plateau stops training. Validation whole-Trial-equal confusion macro-F1 selects weights, breaking exact ties by lower cross entropy. Neither arm reaches the 2000-epoch cap. Traversal is checked from code and epoch counts, not separately recorded batch-index tapes.

Input	Full epochs	Best epoch	Updates	LR reductions
Instantaneous 15D	620	201	140,120	8
Causal 75D	802	423	181,252	8

Table X. Completed classification training: 226 updates per full epoch, eight rate reductions per arm, and minimum-rate plateau stopping. The 32 validation geometries alone select the best epoch, not test scores or real localization.

Model	Macro-F1	Balanced accuracy	Accuracy
TRAIN-majority	0.1427	0.2000	0.5546
Instantaneous 15D	0.4534	0.5432	0.4171
Causal 75D	0.6158	0.6747	0.5609

Table XI. Command-action recognition on 64 held-out virtual geometries. Each Trial's confusion matrix is normalized by eligible frame count before equal-Trial aggregation and calculation of these scores. Imbalanced classes give the majority dummy high accuracy but low macro-F1. Unscored frames, class counts, and every per-Trial confusion matrix remain in the supplement.

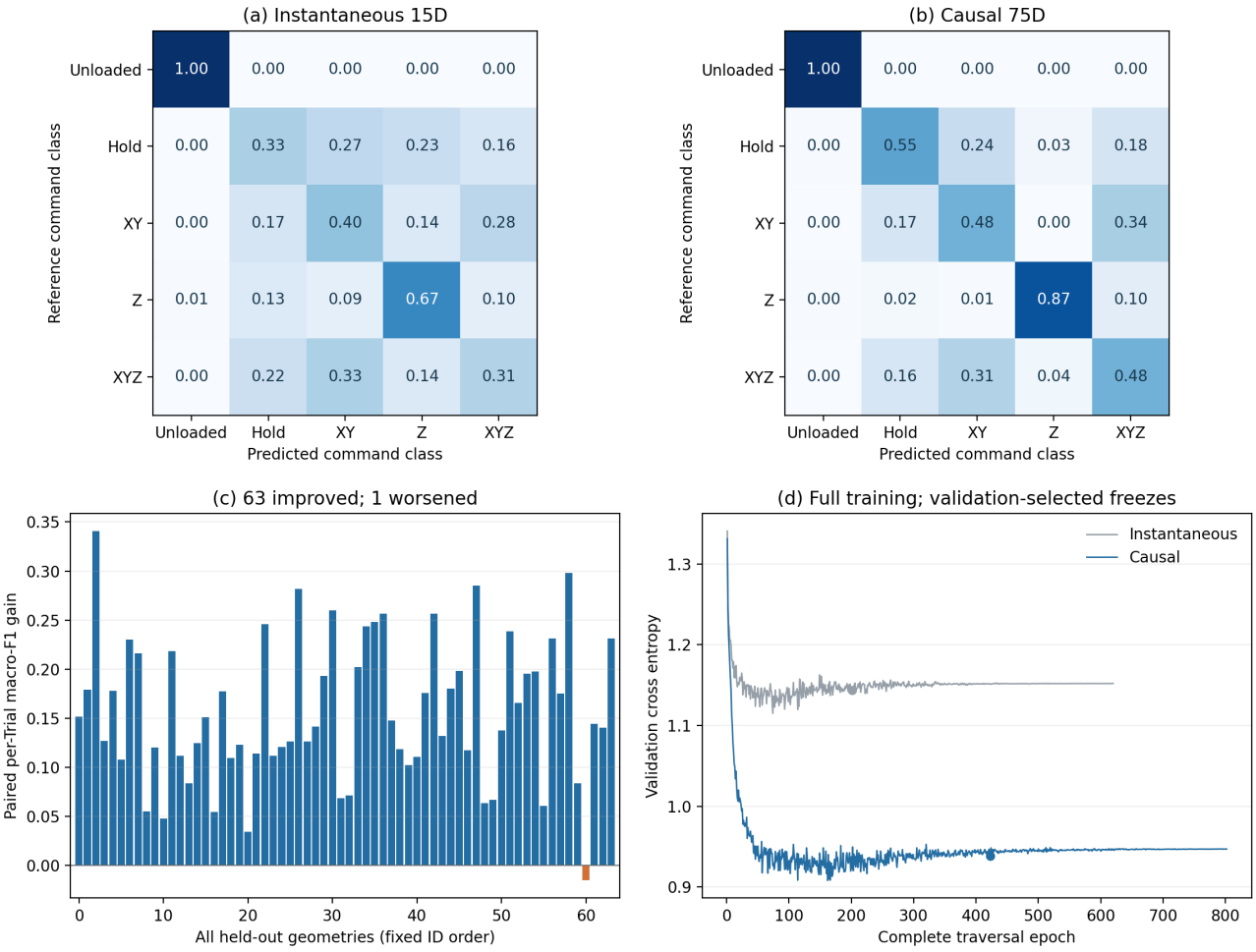


Figure 7. Complete virtual classification comparison: row-normalized whole-Trial-equal confusion matrices (top), all 64 per-Trial macro-F1 differences in fixed ID order (bottom left), and full validation cross-entropy curves with selected weights (bottom right). Unloaded means nominal unloaded, not independently measured physical contact. Input widths differ.

Per-Trial macro-F1 improves on 63 trajectories and worsens on one. The paired mean gain is 0.1562, with a 95% trajectory-bootstrap interval [0.1385, 0.1741] from 10,000 resamples and seed 41091. This estimand differs from subtracting the two aggregated macro-F1 scores in Table XI and excludes training-seed uncertainty. Among 226 fixed class transitions, response requires three consecutive correct frames within 1.5 s. Instantaneous and causal models respond to 99 and 129, with 127 and 97 censored. Observed-only mean delays are 0.340 and 0.315 s; different censoring precludes a claim of faster physical response.

The positive result is a new causal task learned from dynamic virtual magnetic streams, with consistent class benefits on independent geometries within the same generator regime. It does not establish real action recognition or arbitrary-contact recognition, nor does classification improvement imply lower XYZ localization error.

VI Discussion

A Simulation as a research data environment

The clearest simulator-internal result is learnable dynamic coverage from diverse new motion. Matched real/virtual experiments also reproduce the direction of slow-history compensation and causal planar learning. These address different questions: generated data's learning value and reproduction of problem-solving mechanisms. The platform therefore supports rapid motion composition, module testing, and informative real experiment proposals, not just attractive trajectory rendering.

Dwell experiments illustrate theoretical research: observe residual nonmonotonicity, decompose its timescales, formulate a falsifiable input hypothesis, and train matched-size networks rather than stop at a plotted curve. Conditional short-history gains and inconsistent multiscale benefits narrow the next design space. Training and testing from one generator prove learning of a surrogate mapping only. New phenomena require independent real records or validated physical solving; a surrogate cannot validate its own physical correctness.

Source-fitted generators retain device, indenter, and boundary-condition biases. Large virtual training may reinforce these biases and shift coordinates. The 1000-path candidate benefits earlier Yellow replay and one new path's shape and continuity; the ordinary 128-path candidate offers stability with accuracy costs. These are distinct useful directions, not a single verdict on all augmentation. Source raw-error regression and coordinate bias motivate separating coverage limitations, spatial-response mismatch, and candidate-dependent coupling. Domain randomization, independently identified material distributions, and physical surrogates are possible future approaches, not completed results here.

B Depth and safe output require separate checks

After choosing planar weights, candidate-dependent depth evidence is captured for 100-, 500-, and 1000-path depth-branch training. All experiments finish complete traversal and plateau stopping, but no checkpoint passes the original source per-category accuracy/stability protection. We retain frozen baseline depth and do not evaluate unqualified depth weights on the target device. Current benefits are primarily planar; displaying 3D curves is not evidence that every XYZ component was retrained successfully.

For an independently fitted known conical indenter, adding complete vertical indentation/withdrawal history lowers four-family-equal holdout magnetic NRMSE from 51.1% to 40.6%; compound motion still underperforms static fitting. This supports applying axis-separated history to another known geometry with separate coefficients, not zero-shot prediction for arbitrary tips. Appendix B retains all categories without pooling them with the current 5 mm indenter.

C Joint mechanical and magnetic extensions

Cross-tip, tilted, rolling, or multipoint contacts require geometry, orientation, and a shared deformation state in forward inputs. Multiple contacts deform the same layer; adding two independent single-contact magnetic responses is not generally valid. A feasible route is shared mechanics grounded in actual geometry and identified materials, followed by displaced-source fields at fixed Hall nodes and a fast learned surrogate.

The empirical model provides a benchmark and data interface, not proof that fifteen fitted channels uniquely determine a complete spatial field or silicone displacement field. Unmeasured field predictions require held-out measurement locations. Unknown-tip reconstruction requires separate geometries and actions. The present theory and results motivate these questions without counting unqualified physical simulations as validated contributions.

VII Conclusion

We position soft magnetic tactile simulation as a research data environment for discovering and testing improvement directions, not a requirement that every candidate and action improve simultaneously. A source-fitted symmetric axis-separated dynamic surrogate supports continuous interaction, diverse motion generation, and a causal reconstruction loop. Matched real/virtual experiments reproduce the main directions of slow-history compensation and causal planar learning. The sufficiently trained 1000-path candidate substantially improves planar reconstruction on independent virtual geometries and reduces common-valid-frame Yellow XY error from 0.740 to 0.630 mm (14.8%) on existing records excluded from training and checkpoint selection. In two separately acquired same-protocol online runs, the first complex path improves centered XY from 0.267 to 0.189 mm (29.4%) and residual jump rate from 3.91% to 1.90% (51.4%). The second reduces jump rate by 33.0% with a shape-error penalty. A separate ordinary 128-path candidate reduces jumps on all 36 complex real records, with batch means improving by 14.5%, 14.1%, and 6.2%, and reduces dwell fluctuations. These support target-device transfer, action-dependent joint shape/continuity benefits, and temporal stability as distinct useful directions. A new virtual-only causal action task increases held-out-geometry macro-F1 from 0.453 to 0.616, with paired per-Trial gain 0.1562 and improvement on 63 of 64 trajectories. It extends the task value of simulation data but has not demonstrated real transfer. Small simulator-internal denoising gains do not yield additional real-replay benefits and are not counted as positive physical contributions. Original bias and accuracy costs remain visible; centered metrics are not online-origin calibration, and surrogate hypotheses are not physical laws. Simulation expands motion experience, exposes weak conditions, and helps identify promising interventions. Its contribution is reproducible direction discovery and conditional benefit, not a full physical digital twin or universally optimal reconstruction.

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Appendix A Data Roles and Training Provenance

The previous papers and related code use the project repository. Newly prepared simulation, training, and data packages are local upload-preparation artifacts; this address does not imply they have been uploaded. Exact file lists and SHA256 accompany the prepared distribution.

Forward fitting uses 24 source dynamic records D_000086-D_000109 and three static calibrations CAL_000012-CAL_000014; ten source records E_000112-E_000121 select parameters. Targets and declared held-out complex real paths do not participate. Confirmed unloaded baselines, fifteen-channel order, and controller time conventions are indexed with the data.

Role	Whole records	Samples or motions	Use
Source real XY training	32 complete Trials	7217 dynamic samples	Original split retained
Static anchors	Source static constraints	565 samples	Retained in all groups
Source XY validation	Trial 27 grid, 28 eight-direction, 29 circle	3 complete records	Augmentation checkpoint selection
Virtual training	100 to 1000 whole geometries	Four constructions	Seeds from 3701000
Independent virtual test	64 complete paths	Four constructions	Seeds from 3703000
Earlier real replay	7 contact + 2 unloaded per device	18 complete records	Frozen evaluation only

Table A1. Baseline and augmentation data roles. Source files, whole-Trial partitions, initial parameters, generator, virtual directories, and weights have a complete SHA256 index. Regular source validation is not an optimality guarantee for new complex motions.

Virtual TRAIN geometry seed is 3701000 plus path index; TEST seed is 3703000 plus index. A generated validation directory also starts at 3702000, but this sufficiently trained scale study selects checkpoints only on the three original source real validation records. It does not inherit a virtual-selection rule from another experiment. Every augmentation group has one predefined optimizer seed; cross-optimizer-seed confidence intervals are not claimed.

Baseline component	Seed	Full epochs	Best epoch	Updates	Training samples / validation records
XY	7	1023	344	31,713	7,782 / 3 Trials
Z	348	754	117	36,946	12,469 / 12 Trials

Table A2. Frozen baseline training. XY comprises 7217 dynamic samples and 565 anchors; Z comprises 11904 dynamic samples and the same 565 anchors. Both reach minimum-rate plateau stopping and retain their source-validation-best weights. Initial rates are 0.001 and 0.0002; both minima are 0.000001. Augmentation does not add a repeated-old-real-only continuation group. Complete epoch histories and hashes remain in the source index.

Appendix B Depth and Other Known Geometry

Virtual paths	Full epochs	Updates	Final source-objective change	Eight-direction MAE change mm	Qualified weights
100	1139	165,155	-4.54%	+0.0077	None
500	929	457,068	-3.01%	+0.0094	None
1000	750	699,000	-0.89%	+0.0104	None

Table B1. Source depth qualification under new planar weights. These are depth-branch diagnostics on fixed captured evidence, not a new full-XYZ real algorithm evaluation. No scale has a qualified checkpoint, so none enters new target-depth testing.

Holdout motion	Static	Previous history	Added vertical history
Eight directions	51.19%	30.44%	27.19%
Circle	55.10%	42.43%	36.00%
Cross	53.18%	38.26%	35.43%
Compound motion	61.21%	93.33%	63.83%
Four-family equal mean	55.17%	51.11%	40.61%

Table B2. NRMSE for another known conical indenter. Action families are weighted equally and not pooled with the 5 mm indenter. Updated compound motion remains worse than the static reference; every category is retained.

Appendix C Candidate Models and Complete Controls

Positive candidates receive main-text attention without removing the controls. Separate model cards retain the 100/500/1000-path candidates, common64/uniform128/hard128, and raw128/clean128/causal128 with distinct weights and evaluation domains. Targeted expansion mines weak actions only from virtual TRAIN. Its 64-geometry XY error improves from 0.44058 to 0.23287 mm. On the predefined weak-16 subset, it improves over ordinary expansion by 10.19% and 0.02919 mm, below the original stronger criterion requiring both 15% and 0.05 mm. Real localization is not better than ordinary128; it cannot replace the stability candidate on that basis.

In the matched-geometry denoising experiment, clean128 removes only added random noise while retaining dynamics; causal128 filters training streams causally and recaptures full-chain features before sufficient training. With raw virtual test inputs under the eight-condition common mask, they lower XY error beyond raw128 by 2.89% and 4.85%. Applying the same inference filter directly to R373 instead increases error by 2.13%. These are virtual results, with training and inference-filter contributions separated.

Real denoising replay is now complete and independently audited: 48 records from three batches, 192 native causal streams, and 960 metric blocks across four models. Every model receives the same unfiltered real fields and actual B0, dt, resets, events, permission, and depth chain. No clean real ground truth is assumed, and real streams do not refit weights or thresholds. Table C1 contains twelve complex trajectories per batch; SOURCE repeat is another batch of the same source device, not a third device.

Batch / model	Raw XY mm	Centered XY mm	Jump rate	Dwell RMS mm
SOURCE / R373	0.436	0.215	23.23%	0.11195
SOURCE / raw128	0.575	0.253	19.87%	0.10460
SOURCE / clean128	0.581	0.244	25.67%	0.11679
SOURCE / causal128	0.570	0.249	24.19%	0.11314
SOURCE_REPEAT / R373	0.452	0.224	24.56%	0.11706
SOURCE_REPEAT / raw128	0.583	0.264	21.09%	0.10685
SOURCE_REPEAT / clean128	0.590	0.255	26.71%	0.11924
SOURCE_REPEAT / causal128	0.580	0.261	25.36%	0.11641
Yellow / R373	0.521	0.478	46.34%	0.15963
Yellow / raw128	0.679	0.504	43.48%	0.15211
Yellow / clean128	0.733	0.529	49.50%	0.16979
Yellow / causal128	0.707	0.511	48.05%	0.16591

Table C1. Complete denoising-training comparison on real inputs. raw128 and ordinary uniform128 share weights. Accuracy and jumps use this study's four-model common-valid moving frames; dwell RMS is computed per Trial then equally averaged. Centering remains a fitted constant-translation diagnostic, not online calibration. Raw errors and individual coverage remain available. clean128 and causal128 add no real benefit: their jump rates exceed raw128, and neither raw nor centered localization beats R373. Virtual denoising gains do not automatically transfer to real fields.

raw128 lowers jumps on all 36 complex trajectories and lowers dwell fluctuations in all three batches; the two denoising-trained candidates do not reproduce that combination. In R373/raw128/clean128/causal128 order, Z residual jump counts are 1/1/0/1 for SOURCE, 0/0/0/0 for SOURCE repeat, and 1/1/3/3 for Yellow. Complex-trajectory safe references without external clearance hints yield 9/6557 outputs for each SOURCE arm, 0/6682 for each repeat arm, and 0/5179 for each Yellow arm; not every safe record has zero output. SOURCE and Yellow baseline verification matches 97522 online rows; SOURCE repeat is checked against prior frozen replay only. All weights, protocols, complete histories, raw and centered accuracy, coverage, failures, and per-condition outputs are retained. Previously viewed real records remain retrospective frozen evaluation, not new blind tests.

Distributed model cards are in model_results.html and model-results/report.json; corresponding reports, weights, protocols, and histories are in the repository's analysis directories. Production weights and acquisition UI are unchanged. A local preparation package is not already publicly uploaded evidence.

Appendix D Dwell Search and an Unvalidated Hypothesis

For each free candidate, 24 endpoints are selected uniformly in its valid contact segment, totaling 576. Current XYZ is held for another 16 s. A nonmonotonic transient is flagged if residual field norm rises by more than 5% above dwell onset and source-noise-normalized RMS rises by more than 0.5. Sixteen endpoints qualify (2.78%). A post hoc strongest example rises from 62.53 to 117.03 microtesla, peaks at 2.2 s, and decays. It explains the mechanism; prevalence uses all endpoints.

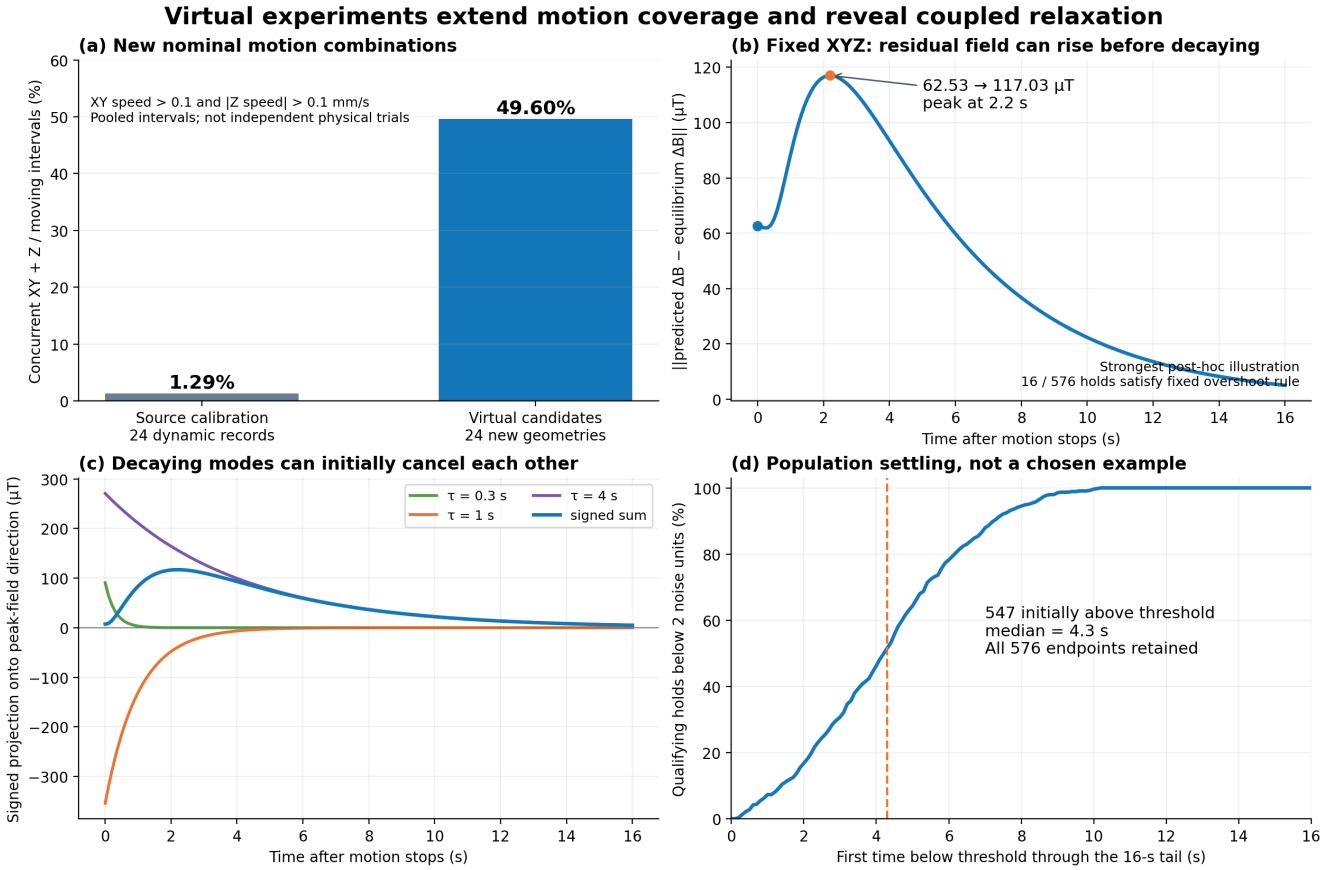


Figure D1. Simulator-internal information from new motion coverage and controlled dwell. Upper left compares coupled XY-Z motion in source training and free virtual candidates. Other panels show 576 endpoints, historical residuals, and one explanatory timescale decomposition. Residual is predicted field minus same-position static equilibrium field, not total field magnitude. Release of cancellation between modes explains the rise and fall; a single-timescale control cannot produce this mechanism. The post hoc example is not representative of all actions.

Analytical dwell continuation and runtime prediction after adding 16 s agree within 1.70e-12 microtesla, attributing the phenomenon to the frozen generator's state structure. Among 547 endpoints with initial residual above two noise units, median first entry below that threshold with persistence through 16 s is 4.3 s. This is not a silicone material constant. Endpoints share 24 correlated paths, so descriptive frequency is not an independent physical occurrence probability. Subsequent quick real experiments fail the original complex-history peak criterion in all six holds (0/6); declared timing-sensitivity checks do not change this outcome. Late same-command-endpoint field differences coexist with unloaded drift and do not validate the original hypothesis. This remains an unvalidated exploration, not a positive physical contribution in the abstract or conclusion.